MOSAIC: Data-Certified Stochastic Release under Structured Deployment Shift

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Code and artifacts: https://github.com/RudraChopra/mosaic-certified-release/tree/mosaic-aaai2027-v6

Abstract

Concept-erasure evaluations rarely certify the public interface that remains after an edit. We present MOSAIC, a software-only release method that either selects a task-specific stochastic channel and decoder or abstains. One simultaneous confidence region over pre-release token laws covers every channel optimized from that table. Labeled target data certify a common-transform plus residual shift class; the unexplained mass is charged exactly to the release channel. We prove finite-sample Bayes-attacker and worst-stratum utility envelopes, global finite-alphabet optimization, and abstention boundaries for unrestricted shift and missing target support. In 1,000 locked safety trials, naive continuum selection releases contract-violating channels on 42.1% of tables; MOSAIC records none. Across 100 registered real-data jobs, MOSAIC releases 20 interfaces with 0/20 diagnostic violations, while a direct target-table certificate releases 36 with 0/36. Under bridge-admitted laws away from those sampled target tables, however, the direct rule violates 16/36 contracts (44.4%); MOSAIC releases the 20 safe cases and abstains on all 16. A 75-job scaling study reaches $K = 64$ tokens and $G = 4$ sources with subsecond median optimization in every cell.

1 Introduction

Representation editing is commonly evaluated by fitting a probe to a frozen embedding (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026), although probe design can change the apparent conclusion (Pimentel et al. 2020). This leaves two deployment gaps: a different downstream rule may recover the attribute, and an edit selected on a noisy validation table may fail after population shift. Both matter when the released token is a public interface.

We study a task-specific finite-token interface. A tokenizer fixed without the certification fold maps an internal representation to $C \in [K]$; a source-blind stochastic channel $M \in \mathrm{Stoch}(K, L)$ releases the only public token Z. Within each task label, the contract bounds the balanced accuracy of the Bayes-optimal source attacker rather than a chosen probe. This isolates conditional source signal from label prevalence. Pre-release shift is a common Markov transform plus bounded source-specific contamination, permitting large source-blind drift while charging the part that can create new source differences.

MOSAIC, Minimax-Optimized Source-Agnostic Invariant Channels, builds an exact attacker envelope from simultaneous multinomial confidence regions. Because the event precedes M, it covers every stochastic channel, including one optimized on the same table. A bridge LP uses labeled target data to learn a common transform and the largest certifiable retained mass; exact transform-and-contamination envelopes then certify source distinguishability and a jointly selected decoder. Finite-alphabet optimization is global, and MOSAIC returns ABSTAIN when no feasible release exists.

The central object is the data-certified bridge: a finite-sample test that turns labeled target observations into a population class on which the chosen interface remains valid. The upstream confidence event then covers the whole post-channel family, so privacy and utility can be optimized jointly without testing each channel. This differs from finite-candidate risk control because the event lies before the channel search. Locked experiments compare this transport certificate with direct target-table certification, plug-in and held-out selection, LTT, deterministic release, official erasers, and an exact population oracle.

2 Related Work

Erasure and universal representations. Concept-erasure methods remove source-associated directions, covariance, or components (Ravfogel et al. 2020, 2022; Belrose et al. 2023; Jourdan et al. 2023; Avitan, Goldberg, and Elazar 2026); interventions and leakage audits explain why a weak probe is not a universal certificate (Elazar et al. 2021; Chowdhury et al. 2025). FARE certifies every downstream classifier for restricted finite encoders, while Fair Normalizing Flows bounds strong adversaries through tractable densities (Jovanovic et al. 2023; Balunović, Ruoss, and Vechev 2022). MOSAIC instead places its confidence event on pre-release token laws, uniformly covering a stochastic post-channel optimized on that table and transferred through an explicit shift class.

Randomized mappings and risk control. Probabilistic preprocessing and fair representation already show the value of stochastic channels (Calmon et al. 2017; Makhdoumi et al.

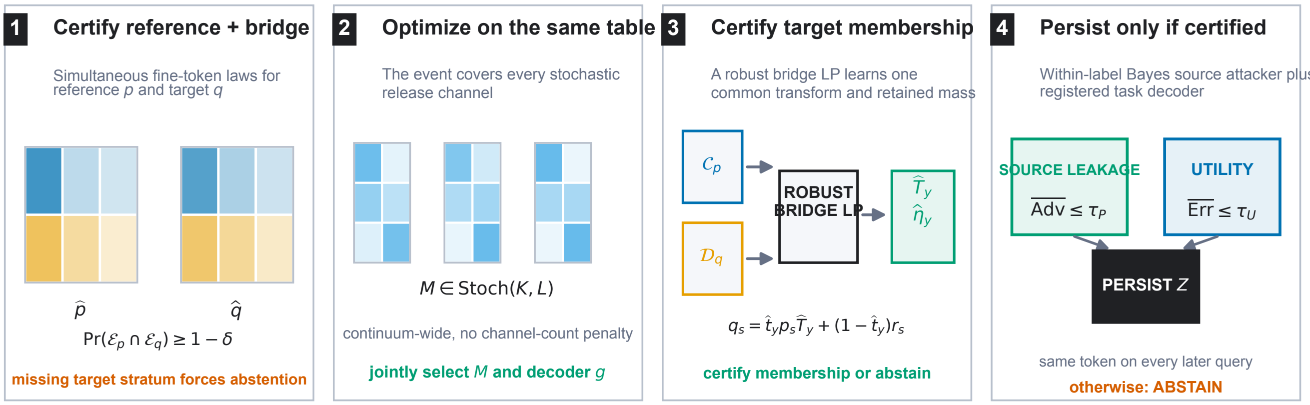

Figure 1: MOSAIC jointly certifies reference and labeled target-bridge token laws, optimizes a continuum of source-blind stochastic channels on that event, and uses a robust bridge LP to certify target membership. The runtime samples one persistent token only when within-label source distinguishability and registered task utility pass; otherwise it abstains. The primary contract is within-label source distinguishability.

2014; Agarwal and Deshpande 2024). Risk-controlling prediction, Learn-Then-Test (LTT), conformal risk control, and Pareto testing certify registered decisions and may abstain (Bates et al. 2021; Angelopoulos et al. 2025, 2024; Laufer-Goldshtein et al. 2023). Robust variants address specified covariate or uncertainty shifts (Tibshirani et al. 2019; Cauchois et al. 2024). These are the closest decision-layer precedents. MOSAIC instead covers an uncountable channel family through one confidence region for its sufficient table, without testing channels separately.

Shift, comparison of experiments, and abstention. Dobrushin coefficients quantify channel contraction (Diaz, Sankar, and Kairouz 2018); fairness and domain-adaptation impossibility results show why unrestricted shift precludes universal guarantees (Lechner et al. 2021; Agarwal et al. 2025; Ben-David et al. 2010). WILDS and subgroup-robustness benchmarks document the practical problem (Koh et al. 2021; Sagawa et al. 2020; Zhou et al. 2021; Sohoni et al. 2020). Bounded-shift fairness transfer and distributional fairness certification give complementary predictor-level robust bounds (Chen et al. 2022; Kang et al. 2022). Common source-blind transformation is Blackwell garbling, while Le Cam deficiency handles approximate experiment comparison (Blackwell 1953; Le Cam 1964; Torgersen 1991); arbitrary residuals follow contamination neighborhoods (Huber 1964). Testable learning asks when target data can validate a shift assumption; recent work studies rejection granularity and tests registered noise models (Klivans, Stavropoulos, and Vasilyan 2024; Chandrasekaran et al. 2025; Patel et al. 2026; Goel et al. 2026). MOSAIC contributes a finite-confidence garbling-and-contamination bridge that composes with the same-table optimizer. Its contract-driven abstention is distinct from predictive-confidence rejection (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017).

3 Problem Setup

Let $S \in [G]$ denote source, $Y \in [J]$ task label, and $C \in [K]$ the fine token. For a fixed label, write $p_{s,y}(c) = \Pr(C = c \mid S = s, Y = y)$. The source-blind release channel M produces $Z \in [L]$. The balanced accuracy of the strongest downstream source attacker within label y is

$$A_y(P, M) = \frac{1}{G} \sum_{z=1}^{L} \max_{s \in [G]} (p_{s,y} M)(z), \tag{1}$$

with normalized advantage $\mathrm{Adv}_G(A) = (GA-1)/(G-1)$. This is chance-normalized and ranges from zero to one. The registered source contract is $\max_y \mathrm{Adv}_G(A_y) \leq \tau_P$.

The main contract concerns source inference within a task label. A joint-law extension covers natural-mixture source inference: define $X = (Y, C)$, estimate $P(X \mid S)$ without balancing labels, and lift the channel as $\widetilde{M}((y, c), z) = M(c, z)$. The same envelope then certifies inference from Z; using output (Y, Z) covers an attacker given the true label as well.

For each supported source–label stratum, the certification table supplies an empirical law $\widehat{p}_{s,y}$ and radius $\epsilon_{s,y}$. The simultaneous confidence set is $\mathcal{C}_{s,y} = \{p \in \Delta_K : \|p - \widehat{p}_{s,y}\|_1 \leq \epsilon_{s,y}\}$. Under conditionally i.i.d. categorical sampling within each registered source–label stratum, the simultaneous event

$$\mathcal{E} = \bigcap_{s,y} \{\|\widehat{p}_{s,y} - p_{s,y}\|_1 \leq \epsilon_{s,y}\} \tag{2}$$

has probability at least $1-\delta$ after a fixed allocation across strata. We use the Weissman multinomial radius (Weissman et al. 2003) in the confirmation, although any valid simultaneous region can replace it. The tokenizer and fine alphabet must be fixed without these observations.

The external law for label y belongs to the structured class

$$q_s = t_y p_s T_y + (1-t_y) r_s, \qquad t_y \geq 1 - \eta_y, \tag{3}$$

where T_y is a source-independent fine-token channel from a registered convex uncertainty set and each residual r_s is

Method	Finite Z	Bayes attacker	Same-table M	Target shift	Shift evidence
FARE (Jovanovic et al. 2023)	yes	yes	no	no	no
Randomized fair reps. (Agarwal and Deshpande 2024)	yes	yes	population	no	no
LTT / risk control (Angelopoulos et al. 2025; Bates et al. 2021)	general	candidate	finite family	optional	model-specific
Blackwell / deficiency (Blackwell 1953; Le Cam 1964)	experiment	all decisions	no	population	assumed
Robust validation (Cauchois et al. 2024)	no	no	no	declared ball	estimated
Testable shift (Klivans, Stavropoulos, and Vasilyan 2024)	no	no	hypothesis	yes	tested
MOSAIC	yes	exact	yes	pre-release	finite-sample

Table 1: Component-level scope of the closest frameworks. "Same-table continuum" means that one confidence event remains valid after optimizing an uncountable stochastic-channel family on that table. The MOSAIC row combines upstream same-table coverage, an exact attacker envelope, and target-shift evidence in one release rule.

arbitrary. Both components pass through the same release channel M. This placement matters: a constant-row software channel cannot leak information that appears before it.

4 Adaptive Within-Label Attacker Certificate

For a bounded vector v, define the exact confidence-set support function

$$\Phi(\widehat{p}, \epsilon; v) = \max_{p \in \Delta_K : \|p - \widehat{p}\|_1 \le \epsilon} p^\top v. \tag{4}$$

It is obtained by moving at most $\epsilon/2$ mass from low to high coordinates of v, or from an equivalent linear-program dual. For attacker assignment $a \in [G]^L$, let $v_{M,a,s}(c) = \sum_z M(c, z)\mathbf{1}\{a(z) = s\}$. We then define

$$\overline{A}(\widehat{P}, M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \Phi(\widehat{p}_s, \epsilon_s; v_{M,a,s}). \tag{5}$$

Theorem 1 (Adaptive exact envelope)**.** *On $\mathcal{E}$, simultaneously for every row-stochastic M, including a channel selected as an arbitrary function of the same table, $A(P, M) \le \overline{A}(\widehat{P}, M)$. For fixed M, the right-hand side is the exact supremum over the product of the stated ℓ_1 confidence sets.*

The proof writes the Bayes attacker as a maximum over deterministic assignments. For a fixed assignment the source confidence sets separate, giving one support function per source; all maxima are finite and commute. The implication is pointwise over the complete stochastic-channel space, which is why adaptive selection needs no channel-count correction.

5 Data-Certified Shift Membership

MOSAIC certifies structured-shift membership from labeled target bridge data. Let a labeled bridge sample from the target population define simultaneous confidence sets $\mathcal{D}_{s,y}$ for its fine-token laws $q_{s,y}$. Write

$$h_{s,y}(w) = \max_{p \in \mathcal{C}_{s,y}} p^\top w, \qquad \underline{q}_{s,y}(z) = \min_{q \in \mathcal{D}_{s,y}} q(z). \tag{6}$$

For each label, the bridge program chooses retained mass t and $W = tT$ by solving

$$\begin{aligned} \max_{t,W} \quad & t \\ \text{s.t.} \quad & W \ge 0, \quad W\mathbf{1} = t\mathbf{1}, \quad 0 \le t \le 1, \\ & h_{s,y}(W_{:z}) \le \underline{q}_{s,y}(z) \quad \forall s, z. \end{aligned} \tag{7}$$

The support functions and coordinate lower bounds have exact LP duals, so Eq. (7) is one linear program rather than a transform search.

Theorem 2 (Adaptive finite-sample bridge)**.** *Suppose the reference and bridge confidence sets jointly cover every $p_{s,y}$ and $q_{s,y}$. Any feasible $(\widehat{t}, \widehat{W})$ with $\widehat{t} > 0$, including the same-table maximizer, defines $\widehat{T} = \widehat{W}/\widehat{t}$ and residual laws satisfying*

$$q_{s,y} = \widehat{t}\, p_{s,y} \widehat{T} + (1 - \widehat{t}) r_{s,y} \quad \forall s. \tag{8}$$

Moreover, Eq. (7) *returns the largest retained mass certified uniformly over the product confidence region.*

Indeed, feasibility and coverage imply $q_{s,y}(z) \ge p_{s,y}^\top \widehat{W}_{:z}$ coordinatewise. The difference is nonnegative and sums to $1 - \widehat{t}$, so normalization constructs $r_{s,y}$. Conversely, robust membership requires exactly these coordinatewise inequalities; maximizing the reference support and minimizing the bridge coordinate yields Eq. (7). Thus the learned singleton transform $\{\widehat{T}_y\}$ and contamination $\widehat{\eta}_y = 1 - \widehat{t}_y$ can be inserted into the certificates below on the same event. If a required bridge source–label stratum is absent, its confidence set is the full simplex, every coordinate lower bound is zero, and Eq. (7) forces $\widehat{t} = 0$.

6 Certification under Pre-Release Shift

We use three names consistently: the *data-certified bridge* tests shift membership, the *transform-exact certificate* enumerates the learned shift geometry, and the *capacity fallback* upper-bounds it when that geometry cannot be enumerated.

Define the arbitrary differential-shift capacity

$$\mathrm{Cap}_G(M) = \frac{1}{G} \max_{a \in [G]^L} \sum_s \max_c v_{M,a,s}(c). \tag{9}$$

This equals $\sup_{r_1,\ldots,r_G} A(R, M)$. For $G = 2$, its normalized value is the Dobrushin coefficient, the largest total variation distance between two rows of M.

For a transform polytope with registered extremes $\mathcal{V}_y = \mathrm{ext}(\mathcal{T}_y)$, define the shifted envelope

$$\begin{aligned} \overline{A}_y^X(M) = \frac{1}{G} \max_{T \in \mathcal{V}_y,\, a \in [G]^L} \sum_s \Big[& \\ (1 - \eta_y)\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; T v_{M,a,s}) & \\ + \eta_y \max_c v_{M,a,s}(c) \Big]. & \end{aligned} \tag{10}$$

Theorem 3 (Transform-exact structured-shift certificate)**.** *On $\mathcal{E}$, every same-table selected M and every external law in Eq.* (3) *satisfy $A_y(Q, M) \leq \overline{A}_y^X(M)$. For fixed M, the right-hand side is the exact supremum over the stated confidence sets, transform polytope, retained masses, and arbitrary residual laws.*

For fixed T, retained mass, and attacker assignment, the source confidence sets and residual simplexes separate. Their support values are respectively $\Phi(\widehat{p}, \epsilon; Tv)$ and $\max_c v(c)$. The latter is no smaller, so the worst retained mass is $1 - \eta_y$. The support objective is convex in T; its maximum over a polytope occurs at an extreme. These finite maxima commute, proving exactness. Since the statement holds pointwise for every M, it survives adaptive channel selection.

When transform extremes cannot be enumerated, a capacity-fallback corollary is available. Define $\rho_T(M) = \max_c \mathrm{TV}((TM)(c, \cdot), M(c, \cdot))$ and $B_y(M) = \min\{\mathrm{Cap}_G(M), \overline{A}_y(\widehat{P}, M) + \rho_{\mathcal{T}_y}(M)\}$. Then

$$A(Q, M) \leq (1 - \eta_y)B_y(M) + \eta_y \,\mathrm{Cap}_G(M). \tag{11}$$

The exact envelope in Eq. (10) is never larger. When $\eta_y = 0$ and $TM = M$, both reduce to the reference certificate; when $\eta_y = 1$, both reduce to exact distribution-free capacity.

Because the fallback separately upper-bounds every law in the same structured class, both transform-exact source and utility values are pointwise no larger than their fallback values. "Never worse" is therefore an implementation check implied by the theorem; experiments measure how often tightness changes the release decision. Proofs are in the supplement.

For decoder $g : [L] \to [J]$, let $u_y(c) = \sum_z M(c, z)\mathbf{1}\{g(z) \neq y\}$. The matching exact conditional error certificate is

$$\begin{aligned}\overline{E}_{s,y}^X(M, g) = \max_{T \in \mathcal{V}_y} \Big[&(1 - \eta_y) \\ &\Phi(\widehat{p}_{s,y}, \epsilon_{s,y}; Tu_y) + \eta_y \max_c u_y(c)\Big].\end{aligned} \tag{12}$$

It is the exact supremum over the same confidence and shift sets. Thus M, g, within-label source distinguishability, and worst-stratum utility can all be selected on one event.

Theorem 4 (No free lunch)**.** $\mathrm{Cap}_G(M) = 1/G$ *if and only if all rows of M are identical. For two sources, if two fine tokens carry different task labels and incur row-wise decoder errors at most e_0 and e_1, then the normalized capacity is at least $1 - e_0 - e_1$.*

Consequently, unrestricted source-specific shift permits exact erasure only by making the public token independent of the fine token. A bounded η_y is therefore an identifiable assumption of the deployment contract, not a hidden technical convenience.

A constant-row channel is also the input-independent, $\varepsilon = 0$ endpoint of local differential privacy; randomized response is its nonconstant noisy relative (Warner 1965). MOSAIC differs by optimizing a task-specific channel under an explicit source-inference contract.

There is a separate identifiability boundary for external auditing. Suppose a required source is absent and let $A_{\mathrm{miss}}(q_0, M) = \sup_{q_1 \in \Delta_K} A((q_0, q_1), M)$.

Theorem 5 (Missing-source audit)**.** *For a release channel M fixed independently of the missing-source law and audit randomness, any protocol with false-certification probability at most δ uniformly over the unobserved law q_1 can claim $A((q_0, q_1), M) \leq \tau < A_{\mathrm{miss}}(q_0, M)$ with probability at most δ.*

The observed audit distribution is identical for every missing law; choosing a simplex maximizer makes every stronger claim a false certification. The same argument makes the utility bound for a missing source–label stratum at least $\max_c(M\ell_y)(c)$. Missing required strata are therefore unestimable, not evidence of safety; the supplement gives the full proof.

7 Global Optimization and Abstention

For fixed decoder, support functions, transformed scores, and residual maxima have linear epigraphs. Enforcing all transform extremes and G^L attacker assignments gives one LP in the KL channel entries; enumerating J^L decoders yields the global finite-alphabet optimum. The capacity fallback uses one mixed-integer LP per decoder. Acceptance requires solver optimality, valid primal residuals, and agreement with separately recomputed certificates.

Release protocol. Fix tokenizers, alphabets, thresholds, split roles, and confidence allocation before viewing certification or bridge folds. Construct simultaneous regions, solve Eq. (7), and enumerate decoder LPs subject to every source constraint. Replay each returned certificate, then select the feasible registered tokenizer with minimum certified error under a fixed tie break. Zero retained mass, missing support, no feasible row, or a failed numerical check returns ABSTAIN; otherwise only (M, g) and Z are released.

With $R = \max_y |\mathcal{V}_y|$, the direct formulation is polynomial in K for fixed (G, J, L, R) but exponential in L through attacker and decoder enumeration. Our constraint-generation path solves a restricted LP, finds the strongest omitted attacker exactly, and adds it until none violates the contract; finite termination follows from the G^L assignment family. Endpoint enumeration, dense channel grids, and independent certificate recomputation check the encoding; the supplement reports scaling.

8 Hash-Locked Evaluation

Synthetic confirmation. Before outcomes, we SHA-256 locked the protocol, code, evaluators, seed exclusions, gates, and theory curve. Two source–label populations, two shift regimes, eight rules, and up to five sample sizes use 1,000 independent tables per cell and familywise $\delta = .05$. A separate population evaluator grades every selected channel, and replay scripts reconstruct every decision, risk, interval, and optimizer check. The primary contrasts test unsafe plug-in selection and safe retention against held-out certification, LTT, and deterministic release; every registered cell remains reportable.

Real bridge confirmation. We registered 100 fresh jobs: 20 untouched seeds on each of Waterbirds, Camelyon17-WILDS, CivilComments-WILDS, BiasBios-Clinical, and

GaitPDB, spanning vision, medical vision, text, and time series. Each job evaluates 13 frozen feature edits from the official INLP, LEACE, R-LACE, TaCo, and MANCE++ implementations; proxies are forbidden. Construction data alone fit a balanced task score and its four-bin tokenizer. Separate reference observations certify the pre-shift table, while the natural external split is divided within each supported source–label stratum into two-thirds labeled bridge data and an untouched one-third diagnostic fold. Neither diagnostics nor their outcomes enter channel or candidate selection.

Target S and Y labels supply the conditional token laws $q_{s,y}$ used by the bridge and the within-label contract. Task-only retraining serves a different objective: it improves a target predictor without constructing these source-conditional laws. Where S is sensitive, collection, consent, and permitted audit use are part of the deployment protocol; unsupported strata trigger abstention.

The familywise level is .05 over 13 candidate tables and both reference and bridge regions. The source-advantage contract is .35; worst-stratum utility thresholds are .30, .35, .40, .45, and .49, with .40 primary. We globally optimize $L = 2$, then reoptimize $L = 4$ only for the best covered $L = 2$ candidate. A strict replay accepts no decision tolerance and an exact-rational audit checks every serialized bridge slack and outward risk bound. The reported strict-v2 outputs use a disclosed structural-zero repair, and the archive retains the original and first-replay artifacts. Locked comparators use the same candidates and folds: capacity transfer, a bridge plug-in without intervals, a validation plug-in that ignores the bridge, and unconditional deployment of the best validation plug-in. All datasets, thresholds, abstentions, and missing-support outcomes remain reportable.

Fresh geographic-shift confirmation. We separately lock five fresh ACSIncome California-to-Texas jobs from Folktables (Ding et al. 2021). Income above $50,000 is the task; binary sex is the audit source and is excluded from the released feature matrix. California supplies the eraser-train and reference folds, while Texas supplies balanced bridge and untouched diagnostic folds with 2,000 observations per source–label stratum. The full 13-candidate official frontier, strict-v2 replay, and independent replay audit are all locked before the five seeds run.

Bridge validity stress. A separate 12,000-table confirmation varies the bridge sample size from 500 to 5,000 per source–label stratum. One population obeys a common transform; two violate the registered model through either source-specific transforms or contamination above the declared budget. Acceptance requires a finite-sample retained-mass lower bound of at least .80. This study tests whether valid membership gains power with data while the two registered violations remain rejected.

9 Audited Results

The preregistered synthetic gates pass. Figure 2 reports every implementable rule in the primary cells; "capacity fallback" is MOSAIC's generic capacity certificate. Independent replay recomputes all 64,000 method receipts, 128,000 population risks, decisions, and aggregate intervals with zero mismatch.

The paired safety difference is 42.1 points (95% bootstrap interval $[39.0, 45.2]$), with 421 plug-in-only violations and no reverse discordance. Across 8,000 capacity-fallback certification tables, no accepted release violates the exact population contract. In the retention cell, capacity-fallback safe deployment exceeds held-out, finite-LTT, and deterministic rates by 53.8, 53.0, and 56.3 points. No deterministic channel is feasible. These finite-run facts align with the $1 - \delta$ guarantee.

A separately locked 5,000-table refinement evaluates exact shift geometry on the same tables. At $n = 125$, transform-exact certification safely deploys on 53.0% versus 0.3% for fallback; at $n = 250$, the rates are 99.9% and 57.8%. The exact objective is no larger on all 5,000 pairs, as theory requires, and all 10,000 method-cell decisions have zero false acceptances. Full cell tables, threshold sensitivity, amendment history, and audits are in the supplement.

Matched finite-family baselines. A separately locked replay places four rules on the same 1,000 tables in each primary scenario. In the retention cell, continuum MOSAIC certifies 579/1,000 deployments (57.9%, exact 95% interval $[54.77, 60.98]$), versus 283/1,000 (28.3%, $[25.53, 31.20]$) for Holm LTT over all 500 fixed channel–decoder pairs. MOSAIC alone deploys on 312 paired tables, LTT alone on 16, and both on 267, a 29.6-point paired difference. The exact McNemar value is 2.28×10^{-72}; it is reported as a descriptive paired statistic. The confidence-region grid and matched deterministic restricted-encoder adaptation both abstain throughout this cell. All four rules have zero observed false acceptance. The deterministic row tests FARE's finite-encoder principle under MOSAIC's shift contract, not FARE's official end-to-end training objective.

Bridge validity and power. Across a separate 12,000-table locked stress study, a correctly specified common transform is accepted on 0%, 31%, 100%, and 100% of tables as the bridge stratum size rises through 500, 1,000, 2,000, and 5,000. Underdeclared contamination has population retained mass $.775 < .80$; a source-specific transform has minimum retained mass .50. Both violating populations are rejected on every table at every sample size. Independent replay finds no failure on the joint confidence event and no invalid false acceptance. The study establishes power against both registered bridge violations and shows the acceptance transition as bridge support grows.

Real data-certified bridge. At the primary error contract .40, strict MOSAIC deploys on 20/100 jobs (20.0%, exact 95% interval $[12.67, 29.18]$), all externally estimable, with 0/20 observed violations (one-sided 95% upper bound 13.91%). The selected official edits comprise six INLP, five LEACE, and nine MANCE++ candidates. Capacity transfer deploys none. The bridge plug-in deploys 47 and violates 7/47 estimable diagnostics (14.9%); validation-only selection deploys 80, including 20 unestimable Camelyon jobs, and violates 18/60 estimable diagnostics (30.0%). Unconditional deployment violates 38/80 estimable jobs (47.5%)

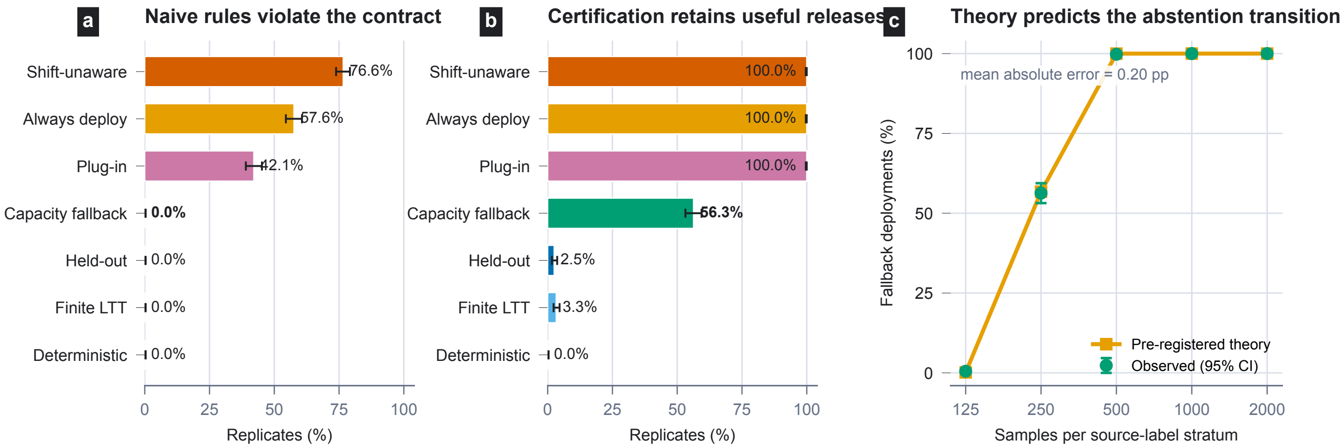

Figure 2: Hash-locked confirmation with 1,000 independent certification tables per cell. Error bars are exact 95% Clopper–Pearson intervals. (a) At the safety boundary, plug-in continuum selection falsely accepts 42.1% of tables; MOSAIC abstains with no false acceptance. Ignoring shift is worse. (b) In the retention cell, the capacity fallback safely deploys on 56.3%, versus 2.5% for held-out certification, 3.3% for finite LTT, and 0% for deterministic MOSAIC. (c) Deployment follows the preregistered theory curve with 0.20 percentage-point mean absolute error. Exact 95% Clopper–Pearson intervals are shown.

and also releases all 20 unestimable jobs. MOSAIC retains 20 safe deployments at every stronger threshold .30–.45 and adds three safe Waterbirds deployments at .49; the supplement enumerates every threshold. These untouched diagnostics show the practical effect of each selection rule alongside the finite-sample bridge guarantee.

The primary releases concentrate on BiasBios-Clinical, where the bridge has strong support and certified median retained mass is .797. The other four benchmarks map the operational boundary of the method: missing source support produces a required abstention, while insufficient joint privacy–utility feasibility produces a contract-based abstention. Waterbirds illustrates the difference between an observed diagnostic and a certificate: its bridge plug-in releases all 20 jobs, whereas the strict procedure declines to release because the joint contract is not certified. The 100 jobs are repeated partitions of five benchmark populations, so results are reported by dataset as well as in aggregate.

What the real ablations remove. The primary rows form a diagnostic ladder rather than unrelated baselines. Capacity transfer keeps simultaneous regions but replaces the learned shift geometry by a worst-case channel-capacity charge; its 0/100 deployments versus MOSAIC's 20/100 show that the exact bridge is decision-relevant. The bridge plug-in retains target data and the fitted transform but removes confidence regions, gaining 27 deployments at the cost of seven observed violations. Validation-only selection removes the target-membership gate, adding 60 deployments but producing 18 estimable violations and 20 unsupported releases. Unconditional deployment additionally removes contract-based abstention and produces 38 violations. Thus intervals, bridge membership, exact geometry, and abstention each change a measured deployment outcome.

Direct target-table comparator. An audited comparator applies the same simultaneous envelope directly to the labeled target bridge table, without making a bridge transport claim. It deploys 36/100 jobs and records 0/36 diagnostic violations, versus 20/100 and 0/20 for strict MOSAIC. This retention gain disappears once the target law moves inside the learned bridge class but outside the sampled-table region. For each of the 36 direct releases, we construct the bridge-admitted law that maximizes its source attacker. The direct rule then violates 16/36 contracts (44.4%, exact 95% interval $[27.9, 61.9]$); MOSAIC releases the other 20, violates none, and abstains on all 16 failures. Every constructed law is an exact member of its stored bridge class and lies outside the direct confidence region. Median structural residual radius is .279, and median distance beyond the direct region is .359 in ℓ_1. The supplement gives the construction, numerical interface utility, and threshold frontier.

Numerical verification and fresh confirmations. The main table uses the strict-v2 arithmetic: deterministic replay matches all 100 receipts, and exact-rational checks verify 1,300 bridge certificates and 1,400 outward release bounds. The archive also preserves the earlier raw and strict-v1 outputs, together with the scoped structural-zero repair that produced v2. A new pre-outcome confirmation on seeds 1220–1224 repeats the corrected pipeline, with 325 candidate and 350 global-optimization replays. It releases all five BiasBios jobs with no diagnostic violation and abstains on the other benchmark jobs at the primary contract. A second pre-outcome confirmation uses the Folktables ACSIncome task (Ding et al. 2021) to transfer from California to Texas. Its strict replay covers 65 official-candidate rows and 70 global optimizations. At the registered .45 sensitivity contract, it releases all five jobs: three R-LACE releases and two degenerate identity interfaces that leave the four-bin score unchanged. All five untouched Texas diagnostics meet the contract. At the .40 primary contract, the best certified errors are .4023–.4145; decomposition attributes .161–.211 to bridge residual

Dataset	Strict	Bridge	Validation	Unconditional	Strict diagnostic	Bridge status
BiasBios-Clinical	20/0	20/0	20/0	20/0	20 estimable	median $t = .797$
Camelyon17-WILDS	0/0	0/0	20/–	20/–	none	missing support
CivilComments-WILDS	0/0	0/0	0/0	20/20	none	no joint feasibility
GaitPDB	0/0	7/7	20/18	20/18	none	missing support
Waterbirds	0/0	20/0	20/0	20/0	none	no joint feasibility

Table 2: Dataset-grouped primary results, 20 seeds per benchmark. Entries are deployments/diagnostic violations among estimable deployments; "–" marks a diagnostic with missing source support. The 20 strict MOSAIC deployments occur on BiasBios-Clinical.

capacity and at most 1.6×10^{-7} to sampling in the active worst stratum. Thus the abstention is caused by the admitted shift class, not a loose multinomial radius.

Alphabet sensitivity. Reoptimizing the best covered candidate with four release tokens is no worse than its two-token solution on all 100 jobs within the registered 10^{-7} solver criterion (largest serialized difference 1.07×10^{-8}), but is strictly better on only five. The real result therefore comes mainly from certifying shift geometry, not from hiding utility in a larger alphabet. It also reinforces the intended scope: MOSAIC is a compact task-specific release interface, not a generally reusable edited embedding. In 75 synthetic jobs spanning $K \in \{4, 8, 16, 32, 64\}$ and $G \in \{2, 3, 4\}$, every .40 contract certifies; the largest cell has .237 median utility slack and .360-second median solve time. Constraint generation uses 4 of 256 possible attacker assignments in a $G = L = 4$ stress point. A real ACS $K = 8$ extension reaches the opposite boundary: retained mass falls to .377–.540, all 13 candidates require a .50-error constant release, and MOSAIC abstains. The supplement reports the full runtime, retention, radius, and sample-size curves.

10 Discussion

Beyond finite candidate testing. LTT and Pareto testing control risk after registering a finite family. MOSAIC certifies the categorical experiment before release: on one covered table, every M and g inherit the same pointwise envelope. The optimizer can then search a continuum without turning internal search points into hypotheses. Tokenizers and erasers are registered statistical objects with their own familywise allocation. The matched 500-pair replay shows the practical result: a structure-aware Holm implementation leaves power on the table when the full channel space is downstream of one confidence event.

Bridge and deployment rule. The bridge measures how much of each target source law is explained by one common transformation of its reference law. Equation (7) ranges over every source-blind $K \times K$ Markov transform and returns the largest jointly supported retained mass t_y; the residual $1 - t_y$ receives the adversarial capacity charge. More bridge data tightens this fit. The protocol registers source and task strata, tokenizer, target population, confidence allocation, and decision thresholds before optimization. A different target population, source definition, or tokenizer defines a new bridge. This gives a clear distinction from target retraining: retraining can improve a target predictor, while MOSAIC certifies a persistent source-blind token interface.

Operational use. MOSAIC is designed for stable software decision interfaces where a channel, a decoder, and a persistence rule can be versioned together. The runtime emits the token when the contract passes and otherwise returns ABSTAIN. The released interface is compact and task-specific, which makes its complete source-inference experiment enumerable and auditable.

11 Guarantee Semantics

The probability $1 - \delta$ is over simultaneous reference and bridge regions. On that event, the conclusion is uniform over optimized channels and every law in the certified shift class. The frozen tokenizer table covers the channel continuum and every decoder without a channel-count correction. Separately learned tokenizers or erasers receive their own registered familywise allocation: MOSAIC pays for statistical objects, not every point searched inside one covered object.

Threat surface. The attacker knows M and may apply any rule to Z. The source-blind software channel is the only public token interface and samples once per immutable item; persistent repeats reveal no more than one Z. If r fresh draws are permitted, certification must use product channel $M^{\otimes r}$; unbounded draws can reveal its row. Transcripts over distinct items require a joint-law certificate. Releasing C, original features, source-dependent metadata, mutable identifiers, or another side channel is outside scope.

12 Deployment Envelope

MOSAIC operates with conditionally i.i.d. reference and target bridge samples that include source and task labels for every required stratum. Source labels are audit data and belong in the associated governance plan. The method releases a compact, task-specific interface; larger alphabets can use structured confidence regions or column generation. The certificate covers within-label source inference from one persistent released token. Multi-item transcripts, prevalence-sensitive analyses, and additional public interfaces use the corresponding joint construction.

13 Conclusion

MOSAIC treats editing as release design. A pre-channel confidence event permits continuum optimization; a target bridge separates common drift from differential contamination. Impossibility results identify required abstention, while locked experiments show unsafe refusals and safe retention under certified shift geometry.

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